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International Mathematics Assessments for Schools

2023 / 2024 JUNIOR DIVISION FIRST ROUND PAPER

Time allowed：75 minutes

When your teacher gives the signal, begin working on the problems.

INSTRUCTION AND INFORMATION

GENERAL

1. Do not open the booklet until told to do so by your teacher.
2. No calculators, slide rules, log tables, math stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 20 multiple-choice questions, each with 5 choices. Choose the most reasonable answer. The last 5 questions require whole number answers between 000 and 999 inclusive. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a mathematics assessment, not a test; do not expect to answer all questions.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are filled in. It is your responsibility that the Answer Sheet is correctly coded.

THE ANSWER SHEET

1. Use only pencils.
2. Record your answers on the reverse side of the Answer Sheet (not on the question paper) by FULLY filling in the circles which correspond to your choices.
3. Your Answer Sheet will be read by a machine. The machine will see all markings even if they are in the wrong places. So please be careful not to doodle or write anything extra on the Answer Sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

INTEGRITY OF THE COMPETITION

The IMAS reserves the right to re-examine students before deciding whether to grant official status to their scores.

2023 / 2024 JUNIOR DIVISION FIRST ROUND PAPER

Questions 1-10, 3 marks each

1. If the square root of a number is between 10 and 11, then the cube root of this number is between which numbers?
（A）3 and 4　（B）4 and 5　（C）5 and 6　（D）6 and 7　（E）7 and 8

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2. Ann bought two types of rice: the first type costs $4 per kg and the second type costs $5 per kg. She mixed the two types of rice in some ratio, then she sold the mixture at $6 per kg, making a 30% profit. What was the ratio (type 1：type 2) of the mixture?
（A）6：11　（B）3：5　（C）5：8　（D）3：4　（E）8：5

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3. A grid has been divided into several non-overlapping rectangles along its grid lines. Exactly one square in each rectangle has been labelled with the area of the rectangle. For example, if the given grid is

	4	6	
2			
		4	

then the dividing method is as follows:

	4	6	
2			
		4	

Now, consider the following grid:

	4					5
		2		4		
	3					
			6			
4		3		★	8	
	1		2			
				5		2

What is the area of the rectangle containing the ★?
（A）3　（B）4　（C）5　（D）6　（E）8

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4. Two towns A and B are at a distance of n km along a river bank. A boat, starting from A and traveling upstream, reaches B in 8 hours. Another boat, traveling the same route, with speed 3 km/h greater than the speed of the first one, covers the distance in 5 hours. What is the value of n?

(A) 40 (B) 60 (C) 80 (D) 100 (E) 120

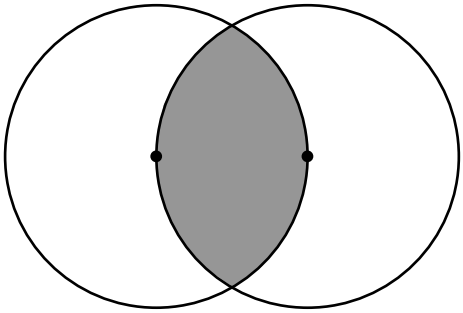
5. The greatest common divisor of the positive integers $3m+4$ and $5m+2$ is equal to d. What is the sum of all possible values of d?

(A) 24 (B) 25 (C) 26 (D) 27 (E) 28

6. What is the remainder when the sum of $1\times 2+2\times 3+3\times 4+...+2022\times 2023$ is divided by 11?

(A) 10 (B) 8 (C) 4 (D) 2 (E) 0

7. In the diagram below, two circles, each of radius 6 cm, are there such that their centres lie on the circumference of the other. What is the area, in cm^2, of the region common to both circles?



(A) $12\pi-9\sqrt{3}$ (B) $24\pi-18\sqrt{3}$ (C) $24\pi-9\sqrt{3}$
(D) $4\pi-\sqrt{3}$ (E) $24\pi-3\sqrt{3}$

8. Eight different digits are written in each cell of a table with 2 rows and 4 columns. Each cell is written exactly one digit. The sum of the digits in any of the two rows is divisible by 7, while the sum of the digits in any of the four columns is an even number. If a and b are the only two digits that are not written in the table, what is the number of possible values for $|a-b|$?

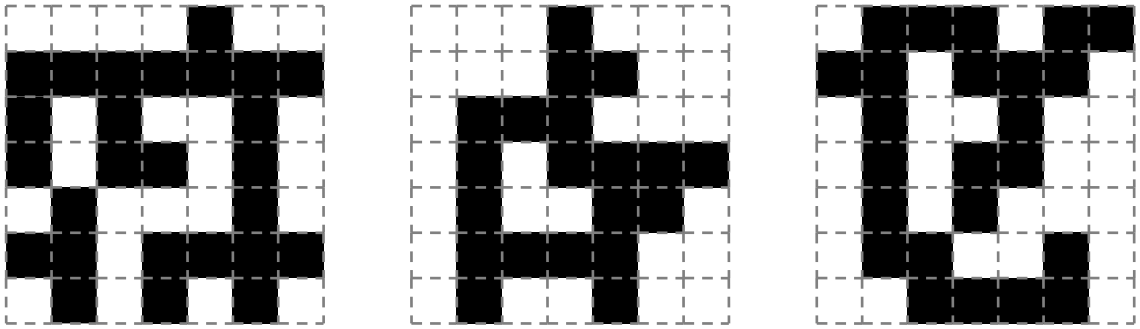
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

9. Several boxes were delivered to a pastry shop. Each box either contain 8 cakes and 6 pastries or 4 cakes and n pastries. The owner of the shop ate 1 cake and 1 pastry and exactly re-distributed the remaining items in boxes of 3 cakes and 2 pastries each. What is the value of n?

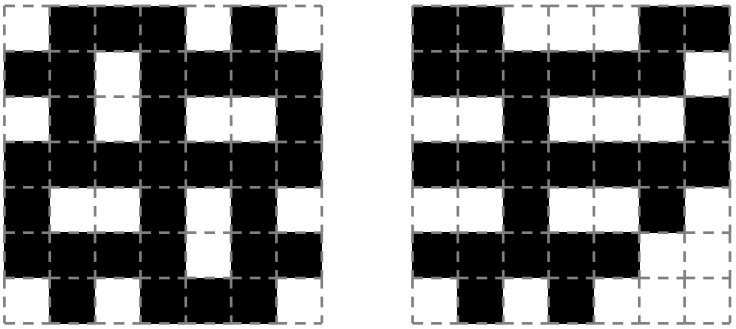
(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

10. Betty paints a 7×7 grid of white squares. At the start, she colours the square in the centre in black. Then, she picks any white square that shares a side with the one and only one other black square and colours it black, repeating this until either she wants to stop or there are no suitable white square exists. Which of the following configurations is a possible diagram that Betty could have coloured?

(A) (B) (C)

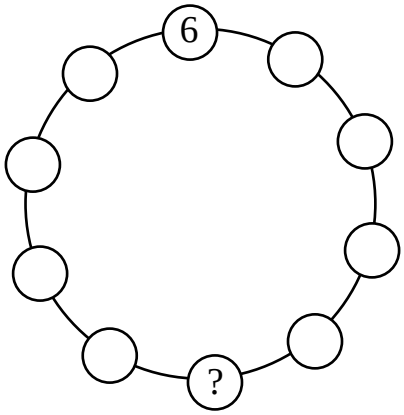


(D) (E)



Questions 11-20, 4 marks each

11. There are knights, which are numbered from 1 to 10, inclusively, are to be seated at a evenly spaced round table. The numbers of any two neighbouring knights differ by at most two. What is the number of the knight sitting directly opposite (marked by "?") to knight number 6?

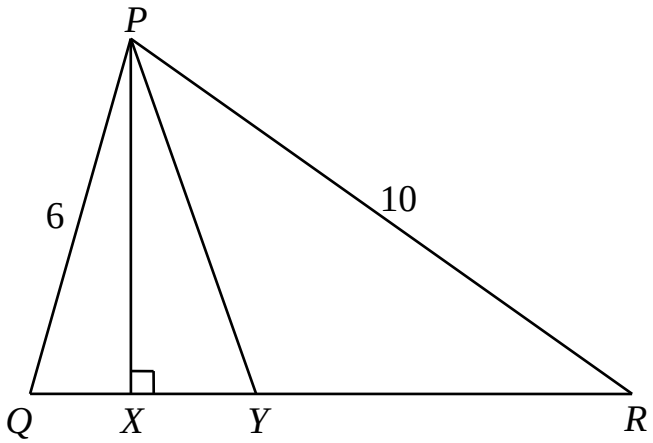


(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

12. There are 2 types of games at a school party, denoted by X and Y. A student gets 7 points for game X and get 11 points for game Y. The students can play x times of game X and y times of game Y. If a student gets exactly 472 points in total, then he can get the special prize. What is the minimum possible value of $x+y$ to get the special prize?

(A) 44 (B) 48 (C) 52 (D) 60 (E) 64

13. Let PQR be an acute triangle with $PQ=6\text{cm}$ and $PR=10\text{cm}$. Let PX and PY be the altitude and the angle bisector from P. If $XY : XR = 1 : 4$, then what is the length, in cm, of QR?

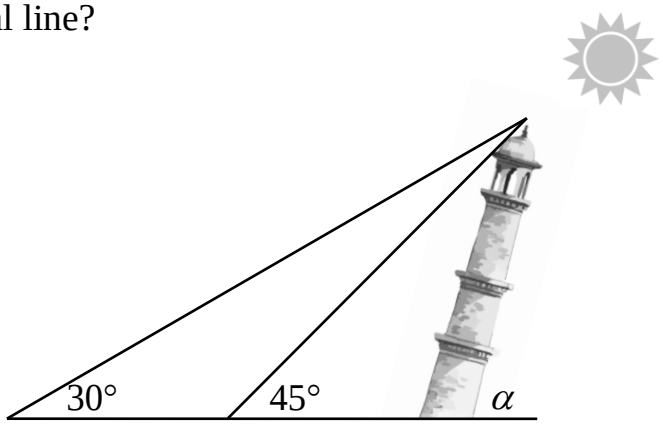


(A) $4\sqrt{3}$ (B) $4\sqrt{6}$ (C) $6\sqrt{6}$ (D) $8\sqrt{3}$ (E) $8\sqrt{6}$

14. If x, y and z are positive integers such that $x+2xy+3xyz=70$, how many possible values of xyz are there?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

15. As shown in the diagram below, when the angle of the sunbeam changes from $45°$ to $30°$, the shadow of a leaning tower doubles in length.
What is the measure of the angle α, in degrees, that the leaning tower makes with the horizontal line?



(A) 60° (B) 65° (C) 70° (D) 75° (E) 80°

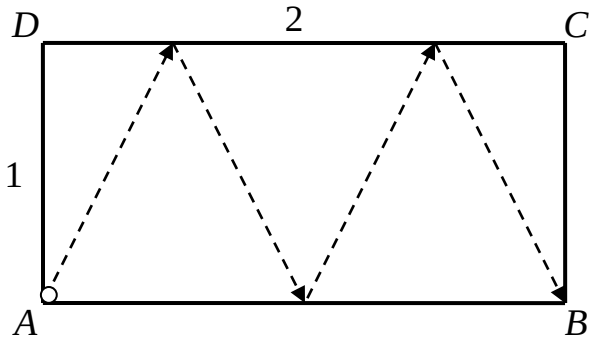
16. The positive real numbers a, b and c satisfy $2a+3b+4c=9$. What is the maximum possible value of $a^4b^3c^2$?

（A）1　（B）2　（C）3　（D）4　（E）5

17. Let $a=\frac{1}{2}(\sqrt{6}+\sqrt{8})$, $b=\sqrt{7}$ and $c=\sqrt[4]{48}$. Which of the inequalities is correct?

（A）$a<b<c$　（B）$a<c<b$　（C）$b<c<a$　（D）$c<b<a$　（E）$c<a<b$

18. A ball is shot from corner A of a 2×1 rectangular billiard table. The ball bounces three times off the walls and then falls into one of the corners of the table. The dashed line is one possible path. What is the longest path it could have possibly travelled?

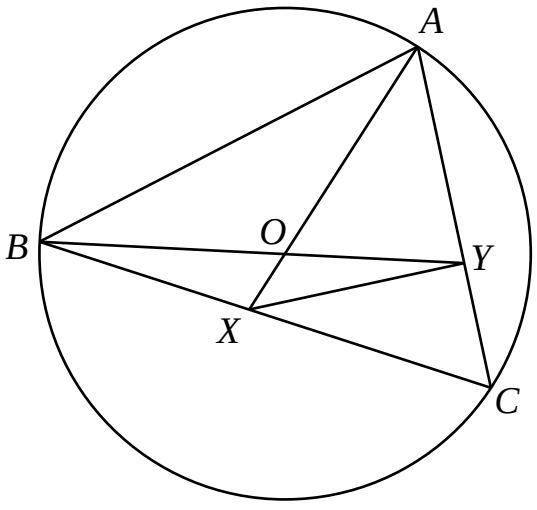


（A）$\sqrt{17}$　（B）$2\sqrt{5}$　（C）$2\sqrt{10}$　（D）$\sqrt{65}$　（E）9

19. From a group of 5 girls and 4 boys, a team is chosen so that it contains at least two girls. In how many ways can this team be chosen?

（A）208　（B）248　（C）416　（D）432　（E）512

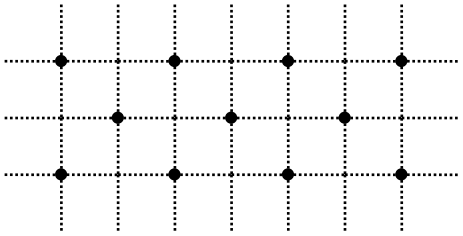
20. Let ABC be an acute triangle with circumcentre O and $\angle ACB=60°$. If AO intersects BC at point X, BO intersects AC at point Y and $XY=7$ cm, what is the length, in cm, of $AY+BX$?



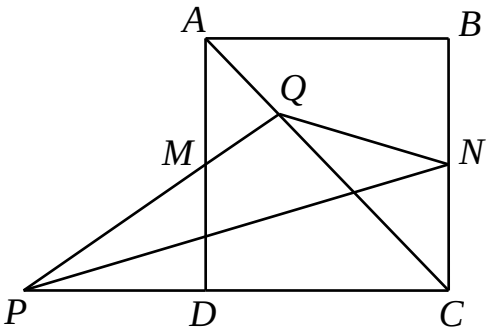
（A）10　（B）11　（C）12　（D）13　（E）14

Questions 21-25, 6 marks each

21. Three points are randomly chosen from the 11 points drawn in the following grid. If $\frac{p}{q}$ (written as an irreducible fraction) is the probability that these three points are the vertices of a non-degenerate triangle, what is the value of $p+q$?



22. Points *M* and *N* are the midpoints of sides *AD* and *BC*, respectively, of rectangle *ABCD*. Point *P* is a point on the extension of *CD* such that $MP = 65\,\text{cm}$. Let *Q* be the point of intersection of *AC* and the extension of *PM*, as shown in the diagram below. It is given that $QM = 26\,\text{cm}$ and $QN = 52\,\text{cm}$. What is the length, in cm, of *NP*?



23. A four-digit number is *good* if it is divisible by the product of the number formed of its first two digits and the number formed of its last two digits. For example, 1352 is a good number since $\frac{1352}{13\times 52} = 2$. What is the positive difference between the greatest and the second greatest good number?

24. A move consists of replacing a given pair (a, b) of integers by one of the pairs:
$$(a+1,\ b-1),\ (a+1,\ b+1),\ (a-1,\ b-1),\ (a-1,\ b+1).$$
Initially, we start with the pair (100, 100). Let *S* be the number of pairs (x, y), where $1 \le x \le 100$ and $1 \le y \le 100$ that can be obtained after a finite number of moves. What is the remainder when *S* is divided by 2023?

25. In how many ways can one arrange two 1's, two 2's, two 3's and two 4's in a row such that there is exactly one "2" between two 1's, exactly one "3" between two 2's and exactly one "4" between two 3's? For example, 12314234 is one possible valid configuration.

* * *